

MH2500 Probability and Introduction to Statistics

Slides for Week 6-2

Topics: Change the order of integration

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Example:

Find the integral,

$$\int_0^1 \int_x^1 e^{y^2} dy dx.$$

According to the limits of integration of the given integral, the region of integration is

$$0 \leq x \leq 1,$$

$$x \leq y \leq 1.$$

We can also describe the region by

$$0 \leq y \leq 1,$$

$$0 \leq x \leq y,$$

the integral with the order changed is

$$\int_0^1 \int_0^y e^{y^2} dx dy.$$

With this new $dx dy$ order, we integrate first with respect to x :

$$\int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 e^{y^2} x \Big|_{x=0}^{x=y} dy = \int_0^1 ye^{y^2} dy.$$

Theorem:

Let X be a continuous random variable with probability density function $f(x)$, then for any real-valued function g ,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx,$$

provided that $\int_{-\infty}^{\infty} |g(x)|f(x)dx$ exists.

We first prove the following proposition

Proposition

For a nonnegative random variable Y ,

$$\mathbb{E}(Y) = \int_0^{\infty} P\{Y > y\}dy.$$

We present a proof when Y is a **continuous random variable** with probability density function f_Y . By $P\{Y > y\} = \int_y^\infty f_Y(x)dx$, we have

$$\int_0^\infty P\{Y > y\}dy = \int_0^\infty \int_y^\infty f_Y(x)dx dy.$$

Interchanging the order of integration in the preceding equation yields

$$\int_0^\infty P\{Y > y\}dy = \int_0^\infty \left(\int_0^x dy \right) f_Y(x)dx = \int_0^\infty x f_Y(x)dx = \mathbb{E}(Y).$$

Proof of the Theorem

$$\begin{aligned}\mathbb{E}(g(X)) &= \int_0^\infty P\{g(X) > y\} dy \\&= \int_0^\infty \int_{x:g(x)>y} f(x) dx dy \\&= \int_{x:g(x)>0} \left(\int_0^{g(x)} dy \right) f(x) dx \\&= \int_{x:g(x)>0} g(x) f(x) dx \\&= \int_{-\infty}^\infty g(x) f(x) dx\end{aligned}$$